

ARE OSTERSUND, Sweden

WMO: 022260

Lat: 63.197N

Lon: 14.486E

Elev: 1234

StdP: 14.05

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.6	(c) -1.9	(d) -11.8	(e) 3.0	(f) -7.3	(g) -6.0	(h) 4.2	(i) -0.9	(j) 28.9	(k) 32.3	(l) 25.7	(m) 29.8	(n) 2.9	(o) 290	(p) 0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.8	(c) 75.6	(d) 60.2	(e) 72.1	(f) 58.7	(g) 68.6	(h) 57.5	(i) 62.8	(j) 71.5	(k) 60.9	(l) 68.5	(m) 59.1	(n) 66.3	(o) 7.9	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	79.2	65.3	57.6	74.1	63.7	55.6	69.0	62.4	28.9	71.1	27.6	68.8	26.3	66.1	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.9	(b) 20.8	(c) 18.5	(e) -13.0	(f) 80.1	(g) 6.1	(h) 3.7	(i) -17.4	(j) 82.7	(k) -21.0	(l) 84.9	(m) -24.5	(n) 86.9	(o) -28.9	(p) 89.6		
(4) 23.9	(5) 20.8	(6) 18.5	(7) -13.4	(8) 65.2	(9) 6.1	(10) 2.7	(11) -17.7	(12) 67.2	(13) -21.3	(14) 68.8	(15) -24.7	(16) 70.4	(17) -29.1	(18) 72.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	21.0	21.5	27.3	36.6	45.3	52.9	58.6	56.5	49.1	39.0	30.2	24.4
	DBStd	15.40	10.95	11.29	8.56	6.46	7.29	6.31	6.15	5.67	5.41	6.33	7.67	10.47
	HDD50	4848	898	799	704	404	185	37	3	8	80	343	595	792
	HDD65	9653	1363	1219	1169	853	610	365	217	271	478	806	1045	1257
	CDD50	701	0	0	0	1	41	125	270	209	52	3	0	0
	CDD65	29	0	0	0	0	0	3	19	7	0	0	0	0
	CDH74	172	0	0	0	0	4	24	111	33	0	0	0	0
	CDH80	13	0	0	0	0	0	0	12	1	0	0	0	0
Wind	WSAvg	8.7	8.7	8.9	9.0	8.1	8.7	9.2	8.2	8.2	9.0	9.3	8.9	8.6
Precipitation	PrecAvg	19.1	1.1	0.8	0.9	1.1	1.4	2.4	3.2	2.6	2.0	1.5	1.3	1.2
	PrecMax	30.0	2.3	3.1	2.0	3.0	3.9	5.2	8.5	6.7	6.6	6.4	2.7	3.3
	PrecMin	11.9	0.1	0.0	0.0	0.2	0.0	0.1	0.0	0.4	0.0	0.4	0.2	0.2
	PrecStd	3.7	0.6	0.6	0.4	0.7	0.8	1.3	1.8	1.3	1.2	1.0	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.9	43.0	47.2	61.6	73.2	77.2	81.7	78.2	66.8	57.6	49.9	44.5
		MCWB	38.0	38.2	39.5	48.1	56.1	57.8	63.6	61.8	57.2	51.2	46.0	41.1
	2%	DB	39.1	39.6	42.8	55.4	68.1	72.9	77.0	73.4	62.8	52.8	45.8	41.1
		MCWB	36.2	36.4	37.8	44.2	53.4	57.4	61.9	60.3	55.0	48.6	43.1	38.1
	5%	DB	37.0	37.6	40.5	50.5	62.7	68.0	73.4	69.6	59.9	50.3	42.2	38.9
		MCWB	34.7	34.9	36.2	42.4	50.7	55.7	59.8	59.0	53.6	46.6	39.7	36.5
	10%	DB	34.5	35.4	38.2	46.6	57.5	64.2	69.7	66.1	57.3	48.1	39.6	36.6
		MCWB	32.6	33.0	34.5	39.8	48.3	53.9	58.8	58.0	52.1	45.1	37.5	34.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	38.8	40.5	48.6	57.5	61.8	66.3	64.7	58.9	52.7	46.3	41.4
		MCDB	40.8	41.7	45.0	59.8	69.3	70.9	76.2	72.6	64.1	56.5	49.4	43.9
	2%	WB	36.4	36.7	38.4	45.4	54.7	58.6	63.6	62.4	56.5	49.7	43.0	38.5
		MCDB	38.5	39.2	42.0	52.9	65.2	68.3	73.1	69.8	61.4	52.3	45.5	40.6
	5%	WB	34.7	35.0	36.6	42.9	52.0	56.7	61.7	60.6	54.5	47.3	39.9	36.5
		MCDB	36.6	37.3	39.9	49.2	61.3	65.9	70.5	66.9	58.8	49.7	41.6	38.4
	10%	WB	32.7	33.0	34.8	40.8	49.2	54.7	59.9	58.7	52.7	45.1	37.8	34.5
		MCDB	34.4	35.0	37.9	45.9	57.1	62.7	67.7	64.8	56.5	47.9	39.5	36.2
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.2	11.7	13.1	14.4	13.8	13.8	12.5	10.4	8.0	7.1	8.6
		MCDBR	9.7	9.4	11.6	19.5	21.9	19.7	19.1	17.4	14.4	10.5	9.2	9.5
		MCWBR	8.3	8.2	8.4	11.2	11.1	9.6	8.1	8.2	8.5	7.9	8.1	8.1
	5% WB	MCDBR	9.5	9.2	10.9	17.1	19.5	17.3	16.9	15.0	12.7	9.8	8.7	9.0
		MCWBR	8.4	8.2	8.3	10.6	10.7	9.3	8.3	8.1	8.4	8.1	7.9	8.0
Clear-Sky Solar Irradiance	taub	0.203	0.264	0.293	0.326	0.332	0.328	0.342	0.332	0.303	0.287	0.213	0.346	
	taud	2.049	2.255	2.332	2.423	2.479	2.516	2.490	2.509	2.575	2.483	2.126	2.790	
	Ebn at Noon	141	219	259	273	281	284	276	269	258	213	135	88	
	Edh at Noon	9	18	25	29	30	29	29	26	20	15	8	6	
All-Sky Solar Radiation	RadAvg	56	245	680	1203	1475	1688	1561	1171	705	305	84	27	
	RadStd	5	31	68	101	147	123	173	99	74	46	14	2	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-221	N/A	N/A	N/A

Nomenclature: See separate page